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Spin-based in-sensor computing magnetic tactile sensor for rapid identification of underwater targets

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Abstract

Reliable underwater tactile sensing technology can promote the effective exploitation of marine resources, which is of significant importance for sustainable human development. Such technology typically requires a combination of waterproofing, intelligence, and high efficiency. Herein, we leverage the in-sensor computing (ISC) neuromorphic device architecture as an innovative platform to construct a diamond nitrogen-vacancy (NV) center-based wireless magnetic tactile sensor (ISC-NVTS). This sensor achieves highly linear force-to-magnetic signal conversion through an array of NdFeB magnetized flexible films and accomplishes ultra-fast magnetic signal detection using diamond NV center magnetic sensing unit arrays operating in a fixed-frequency mode. This wireless connection approach effectively solves the waterproofing issue for tactile electronic devices. Furthermore, we enable adjustable pressure responsivity of the sensing unit array through a microwave multi-parameter NV center electron spin resonance control method, endowing it with the advanced intelligence to execute ISC architecture-based artificial neural network algorithms. Finally, the ISC-NVTS achieved a recognition rate of 95.1% (Random noise 0.1, offline simulation) in a classification task involving five types of marine debris and organisms, with a recognition time of only 6.45 ms. We anticipate that this work will advance the further development of tactile sensors and provide support for the sustainable exploitation and utilization of marine resources.

Introduction

As a treasure trove of Earth's resources, the ocean harbors over 90% of global gas hydrate reserves^{1,2}, 80% of the planet's biodiversity^{3–5}, and 96.5% of its water volume⁶. Against the backdrop of gradual terrestrial resource depletion, the effective development of marine resources holds strategic significance for sustainable human development^{7,8}. Reliable underwater information detection technology is indispensable for achieving this goal. However, the high conductivity and strong scattering characteristics of seawater cause severe performance

attenuation or even failure of efficient terrestrial detection methods like microwaves⁹ and optics¹⁰ in underwater environments. Although acoustic detection has become the mainstream method for underwater exploration due to its low-frequency penetration capability, it is still limited by resolution, making it difficult to obtain sufficiently detailed target information^{11–13}. Therefore, tactile sensing technology based on direct physical contact can serve as an effective means to acquire richer and more accurate marine information. Typically, a reliable underwater tactile sensing technology needs to possess excellent waterproofing, intelligence, and high efficiency. However, its realization remains a challenge in the community, constrained by the waterproof packaging requirements of wired-connected pressure-sensitive units and the serial working mode of the von Neumann architecture-based intelligent sensing system for sensing, memory, and computation^{14–16}.

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Recently, intelligent tactile sensors based on various mechanisms such as piezoresistive, piezoelectric, and capacitive have been successfully developed^{17–19}. Based on the innovative platform of the in-sensor computing (ISC) architecture, they have achieved the elimination of data transmission structures between modules and the parallelization of tactile sensing and computation processes, significantly improving sensor power consumption, response speed, and integration^{20–22}. However, most of these sensors contain conductive force-sensitive structures and wired data connection structures²³, imposing high demands on packaging technology for underwater operation. Moreover, these sensors often require direct contact with the target for high sensitivity, creating a contradiction between waterproofing and sensitivity. As a novel tactile sensing solution, flexible magnetic tactile sensors have demonstrated excellent performance in underwater environments^{24,25}. They employ a separated structure comprising a flexible magnetoelastic force-sensitive film and a magnetic sensor²⁶, allowing the force-sensitive structure to make direct physical contact with underwater targets without any encapsulation, enabling high-sensitivity tactile sensing²⁷. Nevertheless, such sensors face difficulties operating on an ISC architecture platform, mainly due to two limitations: first, traditional Hall and magnetoresistive magnetic sensors lack responsivity adjustability^{28,29}; second, their inherent response speed is relatively slow, severely constraining the parallel sensing-computing advantage inherent to the ISC architecture. Diamond NV color centers offer an opportunity to address these issues. As one of the quantum sensing methods with high technical proficiency, they possess advantages such as high sensitivity, strong process compatibility, and room-temperature operation in magnetic detection. More importantly, NV color centers have rich excitation and regulation methods, which makes it possible to adjust the magnetic field responsiveness. Currently, the method of regulating the spin process of NV color centers using multiple parameters such as microwave frequency/power has been verified to achieve responsivity adjustment of NV color center temperature sensors, thereby realizing temperature field sensors based on the ISC architecture³⁰.

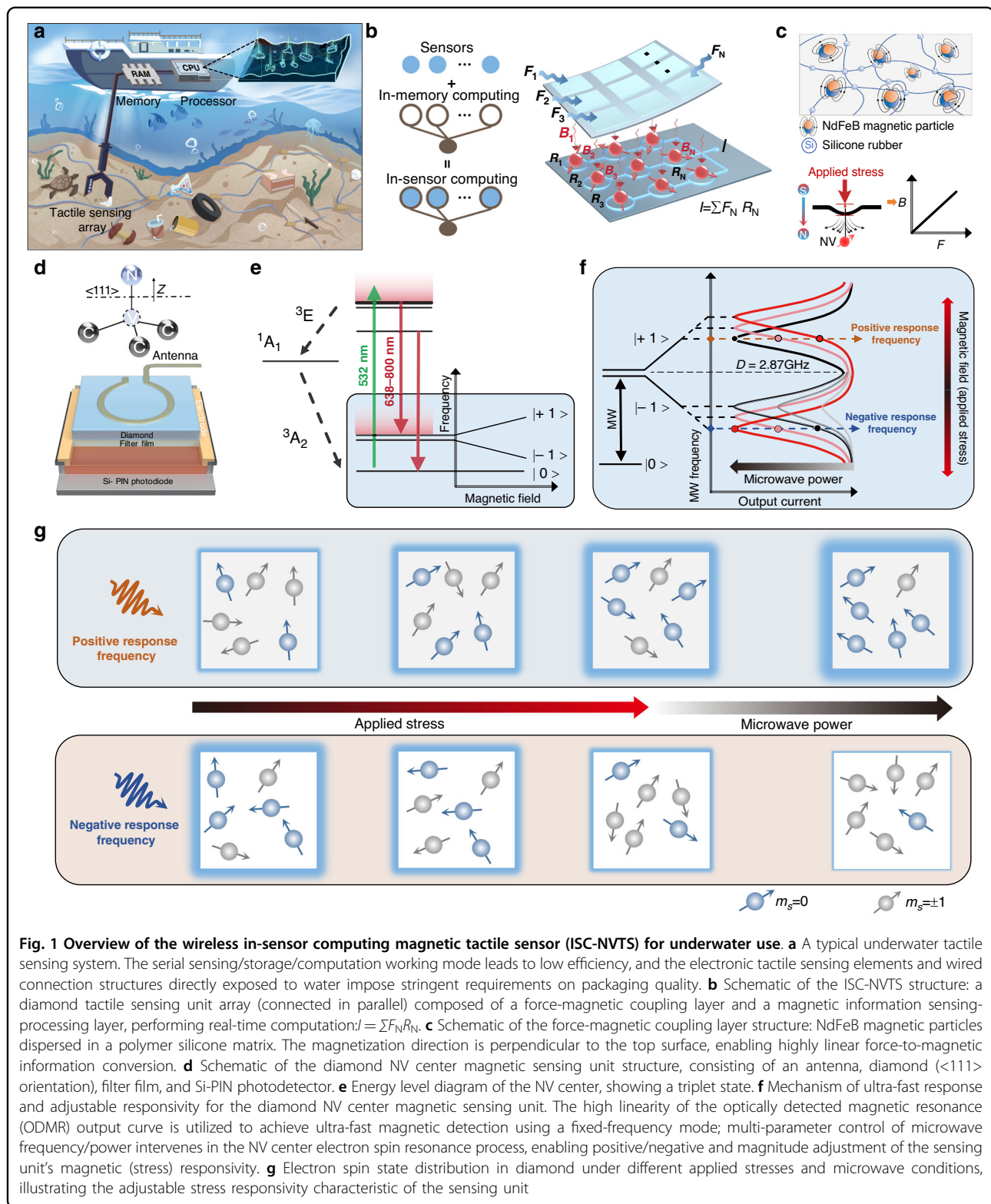
Here, we propose an in-sensor computing neuromorphic architecture-inspired diamond NV center-based wireless magnetic tactile sensor (ISC-NVTS) that utilizes the magnetoelastic effect in soft magnetoelastic materials to achieve underwater tactile perception. The developed ISC-NVTS features a hierarchical structure, including a force-magnetic coupling layer (requiring no encapsulation) and a magnetic information sensing-processing layer. The force-magnetic coupling layer employs an array of NdFeB magnetized flexible films, which can convert tactile stimuli into magnetic field changes

through direct physical contact in an aqueous environment. Leveraging the good penetration of magnetic fields through water, it enables high-fidelity wireless information transmission. The magnetic information sensing-processing layer consists of a diamond NV center magnetic sensing unit array operating in a fixed-frequency mode, enabling ultra-fast magnetic information detection. Concurrently, by intervening in the NV center electron spin resonance process through microwave frequency/power control, adjustable magnetic responsivity is achieved, allowing the NV sensing unit array to implement the ISC architecture. Based on matrix multiplication of magnetic field intensity/responsivity and Kirchhoff's current addition, parallel-mode magnetic information detection and artificial neural network processing are realized. From an application perspective, an ISC-NVTS-based waterproof tactile sensing system for marine object recognition was constructed, achieving a 95.1% recognition rate (Random noise 0.1, offline simulation) in classifying five typical types of marine debris and organisms. This system integrates underwater tactile sensing and computing functions, offering advantages of rapid response, low cost, and high scalability. It provides a reliable technological platform for underwater intelligent tactile scenarios and represents a crucial step towards the sustainable exploitation and development of marine resources.

Results

Wireless in-sensor computing magnetic tactile sensor based on diamond NV centers for underwater use

Current typical underwater tactile sensors (Fig. 1a) adopt the von Neumann architecture, usually comprising separate tactile sensing arrays, memory, and processors. During seabed operations, they sequentially perform tactile information detection, data storage, and advanced computational processing. This serial working mode leads to high system latency, while redundant data transmission between modules results in high system power consumption^{31,32}. More importantly, the tactile sensing array contains conductive structures and complex wired connections, posing significant challenges for packaging technology. Furthermore, paradoxically, high-quality packaging can weaken the information coupling between the target and the tactile sensing unit, severely affecting tactile sensitivity. Therefore, the design principles for underwater tactile sensing systems mainly include two core aspects: first, a tactile sensing mechanism that is unaffected by the aqueous environment; second, an efficient device architecture with a parallel sensing/computation model. Inspired by the ISC neuromorphic architecture, we constructed an ISC-NVTS tactile sensor (Fig. 1b), realizing the fusion of tactile sensing and data processing functions. The ISC-NVTS primarily consists of



a force-magnetic information coupling layer and a magnetic information sensing-processing layer, which together form a diamond tactile sensing unit array. All diamond tactile sensing units are connected in parallel to

form a single output, executing fully connected artificial neural network (ANN) recognition and classification operations based on matrix multiplication of stress (F_N) / responsivity (R_N) and Kirchhoff's current addition

(Supplementary Fig. 1): $I = \Sigma F_N R_N$. ISC-NVTS achieves hardware mapping for the input node scale, node weight matrix, and node interconnection method of artificial neural network models. The internal structure of the ISC-NVTS force-magnetic information coupling layer shows NdFeB magnetic particles dispersed within a polymer silicone matrix (Fig. 1c). After magnetization (direction perpendicular to the top surface), it forms a stable soft magnetic film with a specific magnetic arrangement. During operation, this structural layer can be directly exposed to the aqueous environment for direct physical contact with the target. Based on the giant magnetoelastic effect³³, a highly linear magnetic field conversion can be obtained at the far-field position along the normal line of its center point. The magnetic information sensing-processing layer of the ISC-NVTS is a parallel-connected diamond NV center magnetic sensing unit array, responsible for detecting the stress-coupled magnetic field and executing the ISC architecture ANN recognition/classification algorithm. Figure 1d shows the structure of the integrated diamond NV center magnetic sensing unit based on the optically detected magnetic resonance (ODMR) principle, including an antenna, diamond (<111> orientation), filter film, and Si-PIN photo-detector. As a room-temperature stable solid-state spin system with various processing advantages, the diamond NV center also offers excellent scalability for ISC-NVTS integration.

The diamond NV center is the core of magnetic detection for the ISC-NVTS. It is a lattice defect with an atomic structure where a nitrogen atom replaces a carbon atom in the diamond lattice, forming a vacancy at an adjacent site. The NV center exhibits a 3A_2 spin triplet state ($m_s = 0$ and $m_s = \pm 1$) at room temperature³⁴ (Fig. 1e). When excited by a 532 nm laser, electrons in the $m_s = 0$ state return to the ground state via radiative transitions, emitting photons. In contrast, only a portion of electrons in the $m_s = \pm 1$ states return to the ground state radiatively; another portion relaxes into the metastable 1A_1 state via non-radiative transitions before flipping to the $m_s = 0$ state, a process that does not produce photons. Therefore, fluorescence intensity can be used to infer the spin population distribution within the ensemble of NV centers: a greater number of electrons in the $m_s = 0$ state yields a stronger measured fluorescence signal. When a magnetic field is applied to the diamond NV center, the Zeeman splitting effect occurs, causing the resonance frequencies of the $m_s = \pm 1$ state electrons to shift symmetrically around 2.87 GHz (shift frequency $\Delta f = \gamma * \Delta B$, gyromagnetic ratio $\gamma = 2.8 \text{ MHz/Gs}$)³⁵. Thus, magnetic field information can typically be resolved by frequency scanning to obtain the resonance frequency shift of the $m_s = \pm 1$ state electrons.

The prerequisites for implementing tactile perception on the ISC architecture are: 1. The tactile sensing unit

possesses adjustable responsivity to match different weight connections between nodes in the ANN model^{36,37}; 2. The tactile sensing unit has an ultra-fast response characteristic to maximize the advantages of the parallel sensing/computation mode. Since the force-magnetic information coupling layer of the ISC-NVTS has already converted stress into linear magnetic field information, the requirements here are adjustable responsivity and ultra-fast response for the magnetic sensing unit. As traditional frequency-sweep mode NV center magnetic sensors suffer from high system latency due to frequency sweeping and information resolution, we innovatively propose an ultra-fast magnetic detection method using a fixed microwave frequency, leveraging the high linearity characteristic of the ODMR output curve. Figure 1f shows the frequency-sweep mode ODMR output curves of a diamond NV center magnetic sensing unit responding to external magnetic field changes. When the magnetic field increases, the troughs of the output curve shift symmetrically away from 2.87 GHz (red curve). If the microwave frequency is fixed at the brown dashed line position (positive response frequency), the sensing unit output changes linearly with positive responsivity to magnetic field variations. Conversely, if fixed at the blue dashed line position (negative response frequency), the output changes linearly with negative responsivity. Therefore, the fixed microwave frequency mode diamond NV center magnetic sensing unit can achieve positive/negative responsivity adjustment by controlling the microwave frequency. Increasing the microwave signal power of the sensing unit excites more photons with energy matching the energy difference between the $m_s = 0$ and $m_s = \pm 1$ levels, thereby enhancing the NV center's electron spin resonance effect and promoting more NV center electrons to flip from the $m_s = 0$ state to the $m_s = \pm 1$ state³⁸. Since the fluorescence intensity of the NV center ensemble is proportional to the number of $m_s = 0$ state electrons, increased microwave power macroscopically manifests as a deepening of the trough in the frequency-sweep output curve (from gray to black curve), resulting in increased output responsivity. Thus, adjusting the microwave power of the sensing unit enables control over the magnitude of responsivity. Figure 1g shows the NV center electron spin state distribution in the diamond NV center tactile sensing unit under different microwave and stress conditions. As external stress increases (the magnetic field sensed by the NV center magnetic sensing unit strengthens), the number of NV center $m_s = 0$ state electrons increases (decreases) under the positive (negative) response microwave frequency. As microwave power increases, the difference in the number of electrons in the $m_s = 0$ state of diamond NV color centers caused by the same change in magnetic field will increase (the difference in fluorescence intensity will increase). In summary, by

performing multi-parameter control of microwave frequency/power on the fixed-frequency diamond NV center magnetic sensing unit, we achieved ultra-fast magnetic field response and adjustable responsivity characteristics. Coupling this with the NdFeB soft magnetic film for force-magnetic information conversion yielded an inherently waterproof, ultra-fast responding, and responsivity-adjustable diamond NV center tactile sensing unit, enabling the ISC-NVTS to execute advanced intelligent algorithms based on the ISC architecture.

Implementation and performance of the ISC-NVTS

The force-magnetic information coupling film of the ISC-NVTS is composed of NdFeB magnetic powder compounded with a flexible silicone matrix. The dimensions of the force-magnetic information coupling film for a single tactile sensing unit are 18 mm × 18 mm × 2 mm (Supplementary Fig. 2). To achieve both high magnetic response and suitable flexibility, we systematically studied the influence of magnetic powder mass fraction on the film's comprehensive performance. As the magnetic powder mass fraction in the force-magnetic information coupling film increased from 40 wt% to 70 wt%, the remanent magnetization of the magnetic film continuously increased, reaching a maximum of 50.1 emu/g at 70 wt% (Fig. 2a). However, its coercivity remained essentially stable, indicating that the powder content primarily affects magnetization intensity rather than magnetic domain stability (internal magnetic domains). Additionally, the Young's modulus of the force-magnetic information coupling film increased with higher magnetic powder content. Notably, when the mass fraction changed from 60 wt% to 70 wt% (where magnetic powder agglomeration significantly intensified), the film's Young's modulus jumped from 758 kPa to 1,228 kPa (Fig. 2b), causing a marked reduction in film flexibility. Comparing the properties of films with different ratios, the film with a magnetic powder mass fraction of 60 wt% (Fig. 2c) was selected as the force-magnetic information coupling layer for the ISC-NVTS, providing support for achieving high sensitivity, low latency, and flexible magnetic coupling of external stress information.

The distance between the force-magnetic coupling film and the diamond in the diamond NV center tactile sensing unit is 20 mm (force/magnetic coupling coefficient α is 2.66 N/Gs, Supplementary Fig. 3). Here, ISC-NVTS controls the spacing between adjacent diamond NV color center tactile sensing units to be 20 mm through molds (Supplementary Fig. 4), and the spacing between the diamond and NdFeB film is 20 mm. Applying different loading forces (0 N–0 Gs, 4.97 N–1.87 Gs, 8.82 N–3.32 Gs) causes the Lorentzian troughs of its ODMR frequency-sweep output curve to shift symmetrically around 2.87 GHz. Here, we select

the highly linear regions on both sides of the leftmost Lorentzian trough of the ODMR output curve as the sensing unit's output response curves. Selecting different working frequencies enables responsivity of different polarities (Fig. 2d). Since the ISC-NVTS uses <111>-oriented diamond, after applying a bias magnetic field perpendicular to the diamond's top surface, the outermost Lorentzian trough corresponds to the magnetic field vector component perpendicular to the diamond surface. Therefore, selecting the outermost trough allows the sensing unit to naturally shield magnetic interference from other directions. Determining the sensing unit's working frequency requires balancing linearity and measurement range, i.e., obtaining a higher linearity fitted straight line within the widest possible frequency range on one side of the Lorentzian trough. Through comparative analysis (Supplementary Fig. 5), we finally determined the sensing unit's positive response frequency as 2.81587 GHz. The maximum detectable loading force F_{MAX} at this frequency is 10.72 N [$F_{MAX} = (2 \times \Delta f \times \alpha) / \gamma = 2 \times (2.82151 \text{ GHz} - 2.81587 \text{ GHz}) \times (2.66 \text{ N/Gs}) / (2.8 \text{ MHz/Gs}) = 10.72 \text{ N}$], with a linear correlation coefficient $R^2 = 0.9952$ for the fitted line. The negative response frequency is 2.80755 GHz, with $F_{MAX} = 10.74 \text{ N}$ [$F_{MAX} = (2 \times \Delta f \times \alpha) / \gamma = 2 \times (2.81320 \text{ GHz} - 2.80755 \text{ GHz}) \times (2.66 \text{ N/Gs}) / (2.8 \text{ MHz/Gs}) = 10.74 \text{ N}$] and $R^2 = 0.9958$ for the fitted line (Fig. 2e). Furthermore, Fig. 2d shows that the trough current I_{ct} increases slightly with increasing loading stress. Testing the ODMR output of the sensing unit under different loading stresses (Supplementary Fig. 6) revealed a high linear relationship between I_{ct} and loading force (Fig. 2f, $R^2 = 0.997$). Since the superposition of two linear processes remains linear, this effect does not impact the linearity quality of the diamond NV center tactile sensing unit. The microwave power can controllably intervene in the strength of the electron spin resonance process in the NV center ensemble. The relationship between ΔI (the difference between I_{ct} and the baseline current) and microwave power (0–25 dBm) was calibrated (Fig. 2g), demonstrating the effective modulation of responsivity magnitude by the microwave power parameter. Because the Lorentzian-fitted line of the sensing unit's ODMR output cannot start from zero (not all $m_s = 0$ state electrons in the NV center ensemble can flip to $m_s = \pm 1$, resulting in a baseline fluorescence intensity), a compensation resistor is used to perform a fixed subtraction of the sensing unit's baseline current, allowing its output response curve to start from zero. Finally, we extracted output response curves for the sensing unit under six different microwave power states at both positive and negative (Fig. 2h) response frequencies, all exhibiting good linearity quality ($R^2 \geq 0.992$). At a power of 25 dBm at the positive response microwave frequency, the magnetic

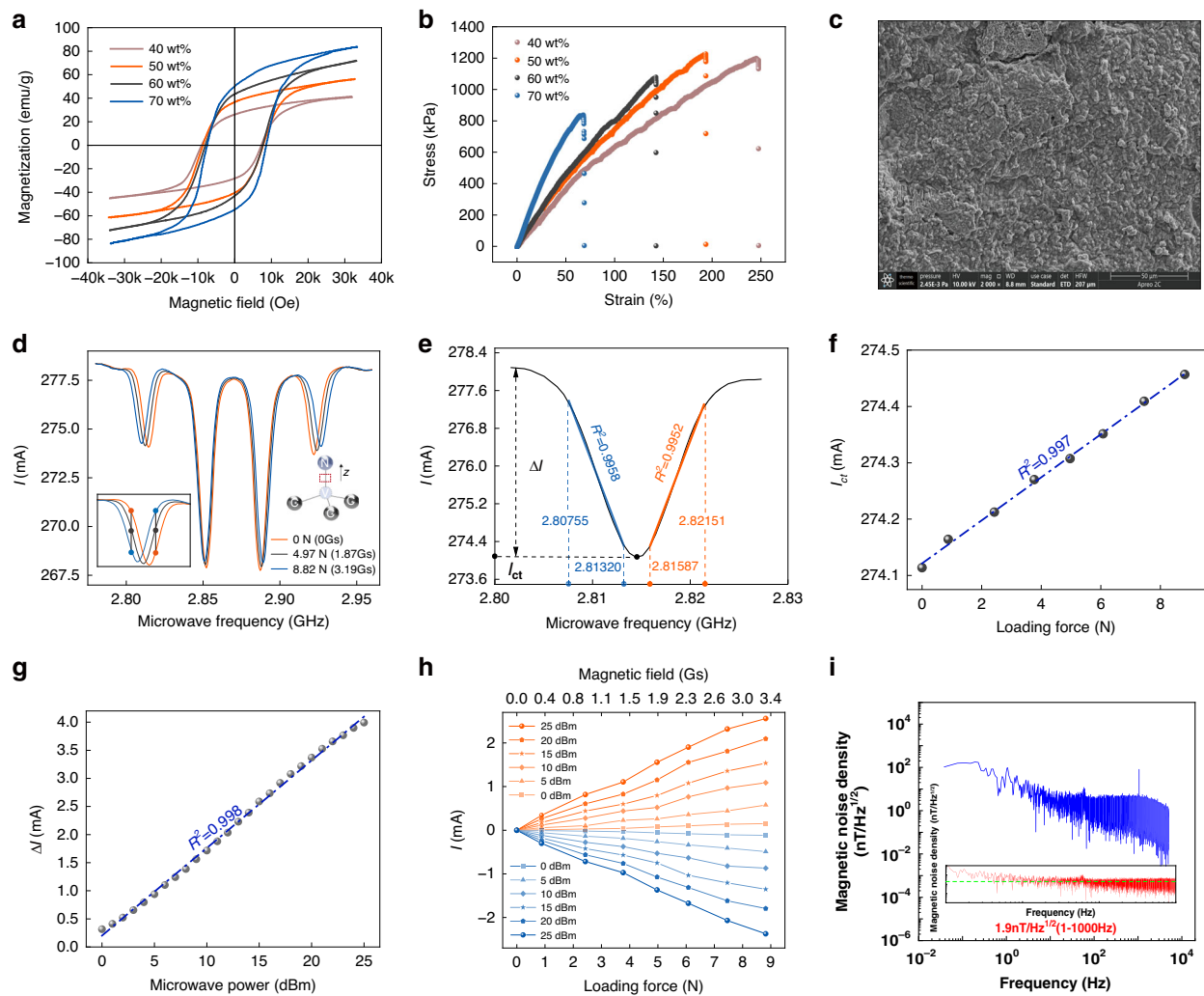
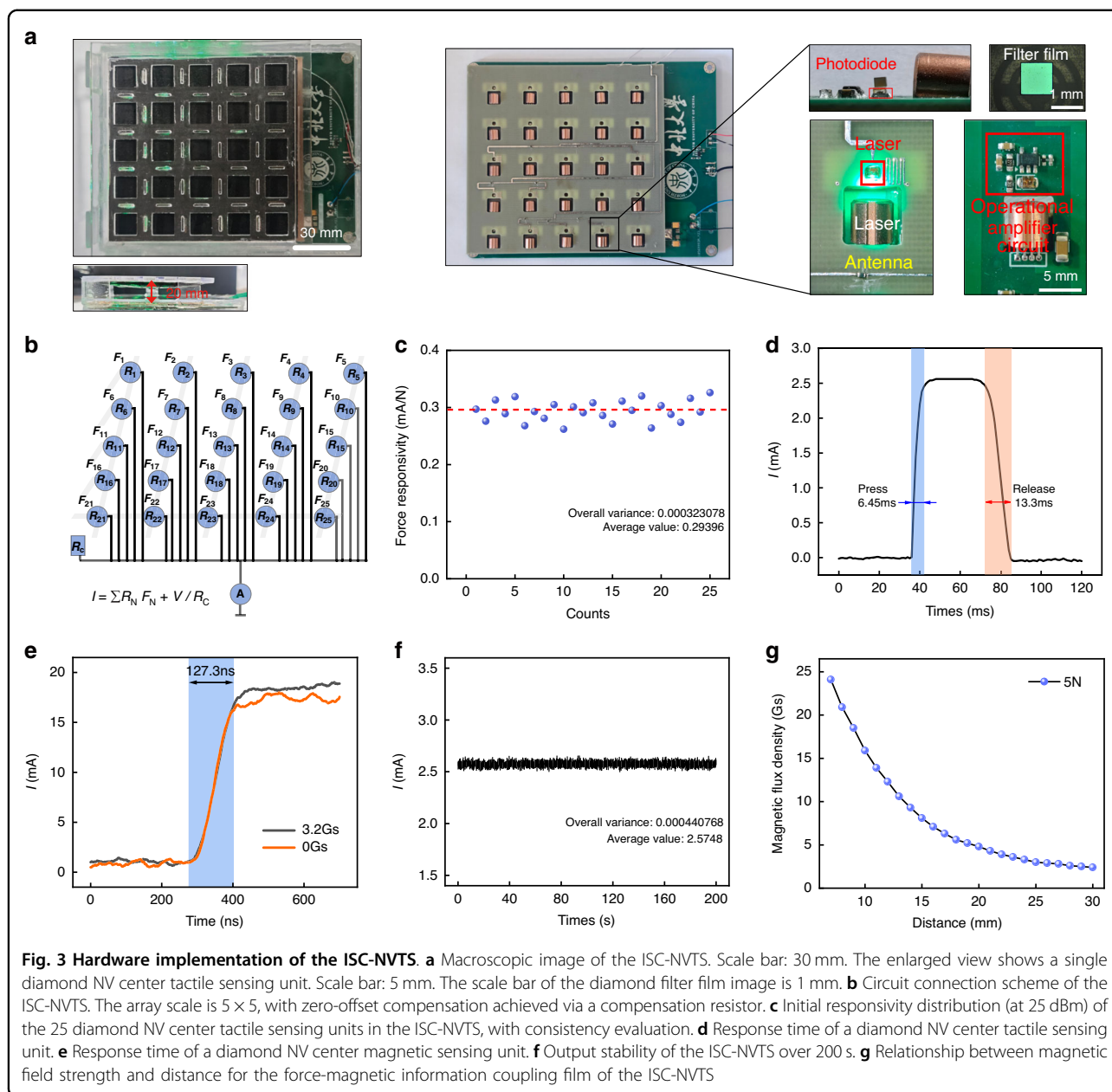


Fig. 2 Performance characterization of the diamond NV center tactile sensing unit. **a** Room-temperature magnetic hysteresis loops of flexible magnetic films with different magnetic particle contents. Saturation magnetization increases significantly with higher particle concentration, while coercivity remains relatively stable. **b** Stress-strain curves of flexible magnetic films with different magnetic particle contents. Higher particle content results in steeper curves and higher Young's modulus, indicating greater rigidity. **c** SEM image of a sample with 60 wt% magnetic particle content. **d** ODMR frequency-sweep outputs of the sensing unit under different loading forces. Selecting different working frequencies enables positive or negative responses. **e** Determination of the optimal working frequency. The highly linear regions on both sides of the Lorentzian trough in the ODMR output curve are selected as the sensor's output response curves. **f** Relationship between the trough current I_{ct} of the leftmost Lorentzian trough and the loading force. **g** Relationship between the current difference ΔI (I_{ct} minus baseline current) and microwave power, demonstrating the adjustable magnitude of the sensing unit's tactile responsivity. **h** Output response curves of the sensing unit at the positive/negative response frequency under different microwave powers. **i** Magnetic field noise spectral density of the sensing unit

field responsivity of the NV color center magnetic sensing layer in the sensing unit is 0.823 mA/Gs (Supplementary Fig. 7), and the noise spectral density is $1.9 \text{ nT/Hz}^{1/2}$ (1–1000 Hz) (Fig. 2i). Its excellent magnetic sensing performance ensures the detection of weak magnetic signals from NdFeB films. The excellent adjustable characteristics of responsivity magnitude and polarity in the diamond NV center tactile sensing unit ensure the feasibility of implementing the ISC architecture in the ISC-NVTS.

The ISC-NVTS was constructed using an array of diamond NV center tactile sensing units. The integrated ISC-

NVTS has a top surface dimension of $100 \text{ mm} \times 100 \text{ mm}$ (Fig. 3a), with the force-magnetic information coupling film thickness being 2 mm and the magnetic information sensing-processing layer thickness being 7.4 mm. Each individual diamond NV center magnetic sensing unit is an integrated structure comprising the diamond, Si-PIN photodetector, laser, and antenna. To enhance the integration of the device, ISC-NVTS employs a spatial optical information coupling approach (Supplementary Fig. 8). Here, ISC-NVTS utilizes transparent resin for the fabrication of the encapsulation structure (Supplementary Fig. 9).



To verify the underwater performance of the current encapsulation structure, hydrostatic pressure simulation tests were conducted in different water depths (Supplementary Fig. 10). Ultimately, the encapsulation structure suffered structural failure when submerged at a depth of 34 meters. In the future, the encapsulation performance can be improved by using non-metallic materials with better fatigue resistance and higher strength. Figure 3b shows its circuit connection scheme and array scale, where 25 sensing units and 1 compensation resistor are connected in parallel to form a single current output channel.

The initial responsivity of the tactile sensing unit array in the ISC-NVTS was calibrated (microwave power:

25 dBm)(Supplementary Fig. 11), showing excellent consistency (overall variance: 0.00032)(Fig. 3c). Since sensing and data processing in the ISC-NVTS occur in parallel, its response time is determined by that of the diamond NV center tactile sensing unit. By applying a transient loading stress, the diamond NV center tactile sensing unit (Fig. 3d) exhibited a rise time of 6.45 ms and a fall time of 13.3 ms, demonstrating excellent response time performance. We also measured the response time of the diamond NV center magnetic sensing unit, with rise time (Fig. 3e) and fall time (Supplementary Fig. 12) being 127.3 ns and 131.4 ns, respectively. Therefore, the delay of the ISC-NVTS is primarily contributed by the force-

Table 1 Comparison of ISC-NVTS response time and similar sensors

Our work	Hu ³³	Hu ⁴¹	Zhou ⁴²	Cui ⁴³	Zhi ⁴⁴
6.45 ms	924 ms	954 ms	583 ms	2258 ms	1049 ms

magnetic information coupling film. The output signal of a diamond NV center tactile sensing unit was tested over 200 s under an 8.82 N loading force (overall variance 0.00044) (Fig. 3f), confirming excellent integration stability. Finally, the relationship between magnetic field strength and distance for the force-magnetic information coupling film of the ISC-NVTS was tested (Fig. 3g). Under a 5 N loading force, the magnetic field strength at a distance of 20 mm was 4.8 Gs, proving excellent wireless signal transmission capability underwater. Tactile sensors based on the ISC architecture possess advantages in power consumption and response time compared to those with the classical von Neumann architecture, due to their parallel sensing/computation working mode and independence from external memory and processors^{39,40}. Due to the integration of arrayed lasers and the requirement for a microwave generator module for operation, the operating power consumption of ISC-NVTS is 8.145 W (see Section S1). The proposed ISC-NVTS shows a clear advantage in response time compared to similar underwater wireless tactile sensors^{33,41–44} (see Section S2) (Table 1). By optimizing the photodetection module and operational amplifier in the diamond NV color center tactile sensing unit, the response time index can be further improved.

Furthermore, it is noteworthy that diamond NV color centers, as a abundant quantum sensing platform, are not only sensitive to magnetic fields but also exhibit excellent response capabilities to temperature fields. We conducted tests on the temperature characteristics of NV color center sensing units (Supplementary Fig. 13) and found that, in addition to the zero-field splitting frequency D undergoing a linear shift with temperature variation (with a shift of -74 kHz/°C), the baseline current value of its ODMR output also exhibits a high linear correlation with temperature. Based on the temperature response characteristics of NV color centers, we propose a three-unit differential magnetic field sensing method (Supplementary Fig. 14). This method can achieve a doubling of magnetic field responsivity on the basis of dynamic compensation for temperature noise, while still possessing responsivity adjustment characteristics and being capable of executing the ISC architecture. This design scheme provides guarantees for the application of ISC-NVTS in marine environments with complex temperature variations.

Using the ISC-NVTS for underwater tactile recognition

Due to a lack of effective regulation, the discharge of various harmful debris during marine resource exploitation has caused irreversible harm to the marine ecosystem^{45,46}. Current primary methods for marine debris removal still rely on human divers, incurring high costs and risks. The ISC-NVTS, with its excellent waterproofing, ultra-fast response, and low-power consumption, can be integrated with an underwater robotic arm to achieve efficient classification of marine debris and organisms (Fig. 4a). Here, a fully connected artificial neural network recognition algorithm (Supplementary Fig. 15) was used for offline training on five typical types of marine debris and organisms (The loading force for the feature pixel points is set to 5 N): battery (9–10.5 mA), reagent bottle (7.5–9 mA), apple (6–7.5 mA), sea turtle (4.5–6 mA), and tire (3–4.5 mA) (Fig. 4b). All the reasoning and training processes are completed on an external processor. Figure 4c shows the loss function decreasing to 7.02 over 8000 training epochs. The inset shows the responsivity distribution of the ISC-NVTS tactile sensing unit array after training, with a compensation current value of 0.962 mA. An underwater tactile recognition experimental system was constructed (Fig. 4d). Models of the five typical debris types and organisms (Supplementary Fig. 16) were randomly placed in a water-filled container. By controlling the robotic arm, the ISC-NVTS made direct contact with the models (The contact force of the feature pixel point is 5 N), and the recognition result was determined based on the output current value. The left side of Fig. 4e shows photos of the recognition targets in an underwater 5×5 grid, the middle shows the corresponding ISC-NVTS output currents, and the right side shows the recognition results judged based on the trained label current values.

To verify the robustness of the ISC-NVTS, 1000 offline recognition tasks were performed with 0.1 random noise, achieving an accuracy of 95.1% (Fig. 4f). In addition, we conducted simulation verification of the recognition function of ISC-NVTS in more complex environments, such as higher noise levels, uneven recognition targets, and a larger number of recognition targets (Supplementary Fig. 17), all demonstrating high recognition rates. Simultaneously, ISC-NVTS exhibits good algorithmic scalability. By increasing the array size of diamond NV color center tactile sensing units and adjusting the interconnection method (Supplementary Fig. 18), an artificial neural network classifier with higher current discrimination can be achieved. The strong robustness and good algorithmic scalability of ISC-NVTS in complex environments provide support for its recognition tasks in complex underwater environments.

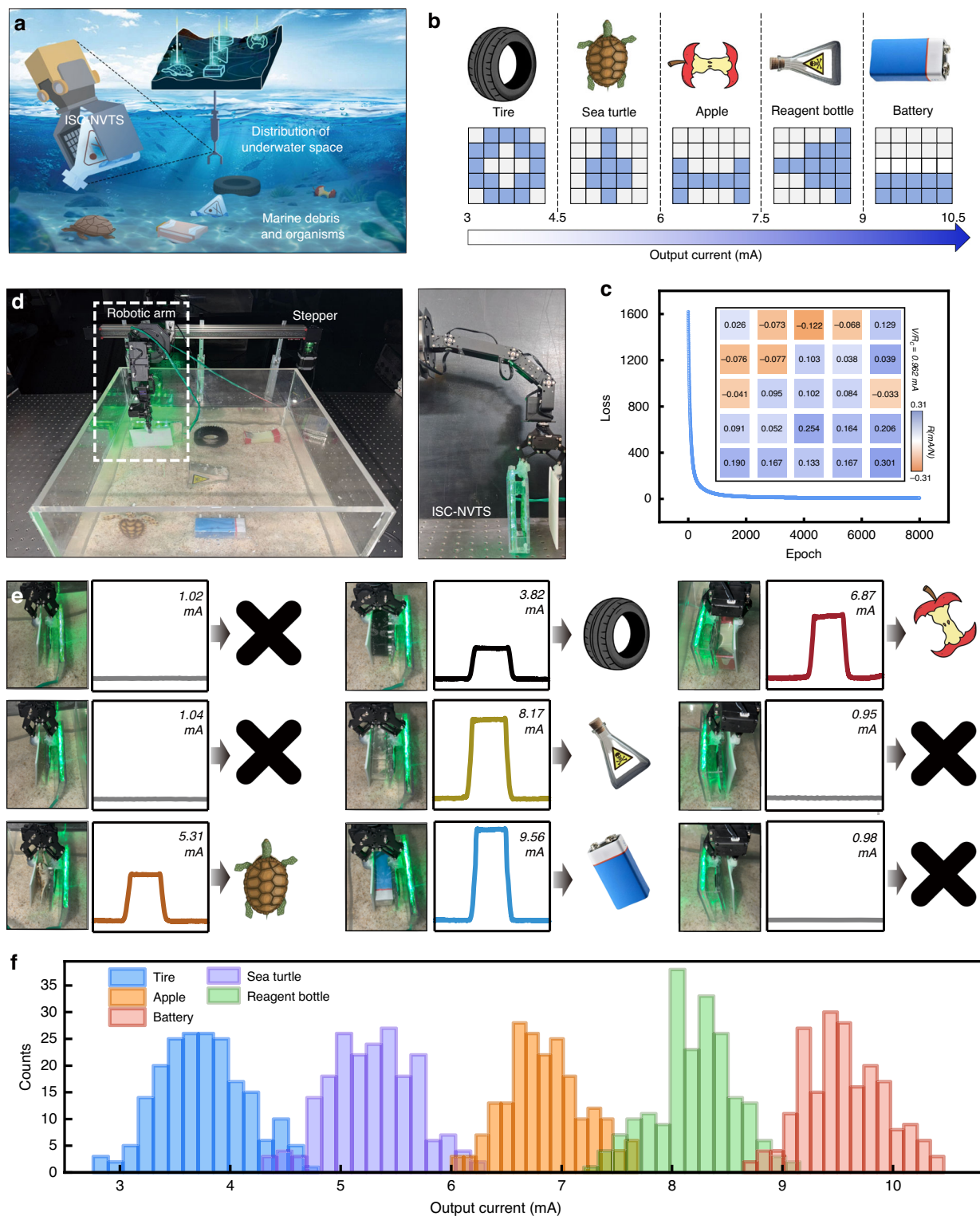


Fig. 4 ISC-NVTS for underwater tactile recognition. **a** Robotic arm integrated with the ISC-NVTS, enabling perception and recognition of marine organisms and debris. **b** Five typical marine organisms and debris, with training feature labels set as: tire (3–4.5 mA), sea turtle (4.5–6 mA), apple (6–7.5 mA), reagent bottle (7.5–9 mA), battery (9–10.5 mA). **c** Loss function variation over 8000 epochs. The inset shows the responsivity distribution obtained from training the ISC-NVTS. **d** Photograph of the constructed underwater tactile recognition experimental system. **e** Left: photos of recognition targets in an underwater 5 × 5 grid. Middle: corresponding ISC-NVTS output currents. Right: recognition results. **f** Distribution of labeled recognition currents for 1000 trials under 0.1 random noise, showing a recognition rate of 95.1%

Conclusion

In summary, we have developed a wireless in-sensor computing magnetic tactile sensor possessing advantages in waterproofing and ultra-fast response. First, our sensor utilizes NdFeB magnetized flexible films to achieve highly linear force-magnetic information coupling. This wireless transmission method allows the tactile sensing film to be directly exposed to the aqueous environment for direct physical contact with target objects, significantly enhancing waterproof performance while maintaining sensitivity. Furthermore, by intervening in the diamond NV center electron spin resonance process through multi-parameter control of microwave frequency/power, we achieved adjustment of both the magnitude and polarity of the responsivity for the diamond NV center tactile sensing unit. This enables it to execute the ISC architecture with a parallel sensing/computation mode, greatly reducing system latency. Finally, we validated that the ISC-NVTS can complete high-accuracy recognition and classification tasks for marine debris and organisms without relying on any external memory or processors.

Discussion

The results of this study provide inspiration for the design and development of highly sensitive, ultrafast-response, low-power, and highly integrated underwater tactile recognition sensors. By coupling force-magnetic information in an underwater environment via a magnetized flexible film, the tactile-sensitive material of the sensor can make direct physical contact with the measured target to maintain high sensitivity, while concurrently reducing the wired connection structures within the sensor to improve waterproof performance. Furthermore, a diamond NV center magnetic sensing unit is employed to collect the coupled magnetic signals. On one hand, this is based on its inherent advantages of ultrafast response, high sensitivity, and integration stability; more importantly, by regulating the spin state of the NV center through multiple parameters such as microwave frequency and power, the polarity and intensity of the magnetic responsivity can be adjusted. This allows it to simulate the weight changes of artificial neural network nodes at the hardware level, thereby achieving a parallel mode for the tactile sensing and recognition processes, which drastically improves the response time of underwater tactile recognition. Notably, the ISC architecture eliminates the storage and processing modules found in classical tactile recognition sensors, thus not only reducing data transmission power consumption but also improving the degree of integration. This sensor, fusing the advantages of quantum precision measurement and artificial intelligence architectures, provides a promising strategy for the design of underwater tactile sensors and offers support for the sustainable development and

utilization of marine resources. Future work will focus on developing tactile sensing units with responsivity storage capabilities to further enhance sensor integration. Additionally, utilizing the diamond NV center as a sensing platform to achieve the detection and neuromorphic processing of physical fields, such as temperature and microwaves, will be pursued to improve the sensor's applicability in complex environments.

Materials and methods

Device fabrication

The force-magnetic information coupling film of the ISC-NVTS was fabricated by compositing NdFeB magnetic particles with a flexible silicone matrix (Ecoflex 00-30 B). During material preparation, the silicone matrix and NdFeB magnetic powder were first mixed in proportion. Fumed silica was added to improve dispersion and adjust viscosity. Subsequently, SE1700 and a catalyst were added to promote cross-linking and curing. After thorough mixing via stirring, casting, leveling, and heat curing, a flexible film with uniform thickness was obtained. Finally, a high-strength pulsed magnetic field of 2 T was applied for vertical magnetization, forming a magnetic film structure with a clear dipole orientation. For the diamond NV magnetic sensing unit array in the ISC-NVTS, the diamond raw material was sourced from Element Six, using the high-temperature high-pressure CD1010 (111) type with a nitrogen content of 100 ppm. After irradiation with a 10 MeV electron beam for 4 hours and annealing at 850°C for 2 hours, the nitrogen-vacancy concentration in the diamond was approximately 0.8 ppm. The filter film selected was an OD3 short-pass filter for 600–800 nm. The laser model is GH15130C8C. The photodetector module utilizes a TEMD5010X01 Si-Pin photodiode, and to reduce power consumption, its output employs a two-stage operational amplifier design. Firstly, an OPA380 operational amplifier (GAIN BANDWIDTH: 90 MHz), which boasts high bandwidth, is used to perform the first amplification of the Si-Pin photodiode's output, enabling it to execute artificial neural network classification algorithms at μA level current. Subsequently, an OPA323 operational amplifier (GAIN BANDWIDTH: 20 MHz), which offers cost advantages, is employed for the second amplification of the operational results, thereby increasing the output amplitude of the device for detection purposes.

The ISC-NVTS antenna (Supplementary Fig. 19) adopts a dual-layer structure with a total thickness of 800 μm , capable of simultaneously radiating microwave signals of two different frequencies. The coupling effect of the array elements of the antenna was simulated (Supplementary Fig. 20). The maximum coupling $S(3,2)$

parameter between adjacent antennas at a frequency of 2.87 GHz is -37.2 dB. The microwave generation module incorporates two customized integrated fixed-frequency microwave generators (Supplementary Fig. 21) with a power consumption of 3.54 W. These generators provide microwave signals of different frequencies to the antenna, enabling the adjustment of responsivity for different polarities of the diamond NV color center tactile sensing unit. The adjustment of the responsivity intensity of the sensing unit is achieved by altering the impedance of the antenna unit, and secondary precise adjustment can be achieved by adjusting the amplification factor of the operational amplifier in the photodetection module.

Experimental setup

The output channels of the ISC-NVTS were all equipped with current-to-voltage conversion circuits. All data were tested using a Tektronix MDO4104C oscilloscope.

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Author contributions

Conceptualization: W.G., H.G., J.T., and J.L.; Methodology: W.G., J.T., and J.L.; Investigation: W.G., H.G., and J.T.; Visualization: W.G., H.G. and J.T.; Funding acquisition: H.G., J.T., and J.L.; Project administration: H.G., J.T., and J.L.; Supervision: H.G., J.T., and J.L.; Writing-original draft: W.G. and H.G.; Writing-review & editing: W.G., H.G., and the other authors.

Data availability

The data generated in this study are included in the manuscript and supplemental information and will be made available from the lead contact upon reasonable request.

Competing interests

The authors declare no competing interests.

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